MUHAMMAD IRFAN BIN RIZAL

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SUMMARY

Cloud Computing & Application student at Universiti Teknikal Malaysia Melaka with hands-on expertise in AWS cloud security, network infrastructure, and full-stack development. Holder of AWS Academy Cloud Security Builder, Oracle Cloud Infrastructure/Fusion AI Agent Studio, and multiple Cisco cybersecurity credentials. 12+ end-to-end projects spanning privacy-first messaging, autonomous cloud security response, multi-cloud disaster recovery, zero-trust data governance, and CSPM automation. Gold Medalist at i-CPROM 2023 (UiTM Perlis) for IT innovation, with a 6-month IT internship at Port Klang Cruise Terminal and an active faculty technical support role.

EDUCATION

Universiti Teknikal Malaysia Melaka (UTeM) *2024 – Present*
Bachelor of Technology in Cloud Computing and Application (Honours) · CGPA: 3.21 | Melaka, Malaysia

Kolej Vokasional Besut *2022 – 2024*
Diploma of Information Technology • CGPA: 3.55 • MUET: Band 3.5 | Terengganu, Malaysia

Kolej Vokasional Besut *2020 – 2022*
Sijil Vokasional Malaysia — Computer & Network Services (Level 3) • CGPA: 3.52 | Terengganu, Malaysia

EXPERIENCE

Fakulti Teknologi Maklumat dan Komunikasi (FTMK), UTeM *Jan 2025 – Present*
Technical Associate (Rakan Teknikal) | Melaka, Malaysia

- Delivered rapid on-site hardware and software troubleshooting for 200+ faculty and student users across laboratory environments.
- Configured projectors, sound systems, and live-streaming equipment for 10+ hybrid faculty events with zero technical failures.
- Served as official event photographer and videographer; produced and edited promotional content for faculty social media channels.
- Supported faculty digital infrastructure including computer lab maintenance and network connectivity checks.

Port Klang Cruise Terminal Sdn Bhd *Apr 2024 – Sep 2024*
IT Intern | Port Klang, Selangor

- Executed routine system health checks and hardware/software verification across 50+ enterprise workstations, significantly reducing operational downtime.
- Assisted lead engineer in upgrading 20+ PoE switches and optimizing LAN/WAN infrastructure, improving overall network stability.
- Performed preventive maintenance and troubleshooting for 30+ CCTV cameras, ensuring 100% continuous surveillance coverage.
- Audited 200+ IT assets and implemented a lifecycle tracking system, achieving 100% documentation compliance.
- Prepared technical quotations for network components and evaluated vendor proposals for cost-effectiveness.

KEY PROJECTS

PanTalk — Privacy-First Messaging Platform (Current)
React, Node.js, Socket.IO, PostgreSQL, AWS

- Developing a secure real-time messaging platform with private messaging, temporary identities, QR-based connections, disappearing messages, group collaboration, media sharing, and privacy controls, deployed on AWS.

Threat Nexus XDR — Final Year Project (FYP): Cloud-Native Autonomous Security Response
AWS EC2, Lambda, DynamoDB, EventBridge, Flask, Chart.js

- Designed a cloud-native XDR platform combining a Cowrie SSH honeypot, a Lambda-based detection engine (10 attack rules), and automated IP blocking.
- Built real-time Telegram/SNS alerting and a Flask + Chart.js dashboard; validated detections via Kali Linux attack simulation.

Omniverse — Multi-Cloud Disaster Recovery (SRE Showcase)
Terraform, AWS EKS, GCP GKE, Python/Boto3, React & WebGL, Solidity

- Automated cross-cloud failover between AWS EKS and GCP GKE with Terraform-managed DR and a Python agent that reacts to SNS alerts to re-route traffic.
- Added chaos-engineering fault injection and an immutable Web3 audit trail logging failover events on-chain.

Cloud Security Posture Audit — Production-Grade CSPM Framework
Terraform, Prowler, Python, CIS Benchmarks, GitHub Actions

- Built a CIS-aligned CSPM framework combining Prowler with a custom compliance checker over Terraform-hardened infrastructure.
- Implemented CI/CD auto-remediation via GitHub Actions, lifting measured compliance from 42% to 87%.

Cloud Data Governance — Zero-Trust Engine for AWS S3 Data Lakes
Python, Terraform, AWS, Next.js

- Engineered auto-classification of PII into security tiers with redaction of sensitive data across S3 data lakes.
- Deployed honeytokens and automated quarantine workflows for compromised user accounts.

AEGIS — Cloud-Native SOC Dashboard
React, Three.js, AWS Amplify, Python (Boto3)

- Built an interactive 3D threat-globe visualization backed by AWS Amplify, with automated AWS remediation actions triggered via Boto3.

Niyyah — Islamic Super App for Muslim Malaysians
React, Next.js, Supabase, TailwindCSS

- Developed a production-grade fullstack app for mualaf (new converts) and warga emas (senior citizens): Al-Quran with audio, JAKIM-integrated prayer times, Muallaf Hub, and Zakat Calculator.

SmartChef — Serverless Recipe Recommendation Engine
AWS Lambda, API Gateway, DynamoDB, Python, Next.js

- Architected a serverless recipe engine on AWS with real-time market cost predictions and nutritional breakdowns via a Python backend.

FunTechPay — Neo-Banking Proof of Concept
React, FastAPI, DynamoDB, Terraform, JWT

- Built a 3D interactive banking dashboard with AI-driven spending insights, a split-bill QR engine, and JWT + TAC-secured transactions.

LEADERSHIP & INVOLVEMENT

- Gold Medalist — i-CPROM 2023 (UiTM Perlis), Intl. IT Innovation Competition
- Protocol & Technical Committee — Program CEO Campus, UTeM (2025): VIP protocol & AV setups
- Safety Secretariat, FTMK Programs (2025)
- Handball Representative, FTMK SAF Ke-19 (2025)
- Participant — Cloud Challenge DTC (2026) & UniPlay (2026)
- Chairman — Mobile Legends Tournament, Volleyball Championship & iTech Club, KV Besut (2024)
- Technical Associate (Rakan Teknikal), FTMK UTeM (2025 – Present)
- Security Committee Member, Career Path Club (2025)
- Volunteer, TVET State & National Programs (2023 – 2025)

CERTIFICATIONS & PROFESSIONAL TRAINING

- AWS Academy Cloud Security Builder (2026)
- Oracle Fusion AI Agent Studio Certified Foundations Associate (2026)
- Python Programming with Red Hat (AD141-RHA) (2026)
- Cisco Network Defense (2025)
- Endpoint Security (2024)
- Ethical Hacker (2023)
- Google Cloud Computing Foundations Certificate (2026)
- Google Workspace Tools & Data Analysis (2025)
- MyDIGITAL & Intel: AI Appreciate & Aware (2024)
- AWS Academy Cloud Security Foundations (2026)
- Oracle Cloud Infrastructure 2025 Certified Foundations Associate (2026)
- Introduction to Microsoft Azure Cloud Services (Coursera, 2026)
- AWS Academy Cloud Foundations (2025)
- Junior Cybersecurity Analyst Career Path (2024)
- CCNA: Introduction to Networks (2023)
- Cyber Threat Management (2023)
- RunCloud: Cloud Computing Fundamentals (2024)
- JPK: SKM Computer Network Services Level 3 (2023)

SKILLS & INTERESTS

Cloud & Infrastructure: AWS (Lambda, API Gateway, DynamoDB, IAM, S3, EC2, EKS, EventBridge, GuardDuty, CloudTrail, Security Hub, Amplify), Google Cloud (GKE, Compute Engine, Cloud Storage, IAM), Microsoft Azure (foundational), OpenStack, Docker, Terraform, Serverless, Virtualization
Security: Network Defense, Ethical Hacking, XDR, CSPM (Prowler, CIS Benchmarks), Zero-Trust Data Governance, Endpoint Security, Cyber Threat Management, Phishing Detection, SIEM/SOAR (foundational)
Networking: CCNA, LAN/WAN, PoE Switches, Load Balancing, VPN, Firewall Configuration, Intrusion Prevention
Programming: Python (Flask, FastAPI, Boto3), JavaScript/TypeScript, React, Next.js, C++, PHP, HTML/CSS, TailwindCSS, Chart.js, Three.js, Solidity, SQL, MySQL
IT Operations & Tools: Linux & Windows OS, Server Maintenance, CCTV Systems, IT Asset Auditing, AV Systems, Git, GitHub Actions, Docker, Supabase, RunCloud, Google Workspace, VS Code
Language / Interests: Malay (Native), English (Proficient - MUET Band 3.5) | Cloud security research, Islamic fintech, esports event management, photography & videography